

WHIDBEY ISLAND NAS, WA, USA

WMO: 720749

Lat: 48.350N

Lon: 122.667W

Elev: 47

StdP: 14.67

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24255

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.7	27.7	11.6	9.9	29.3	16.9	12.9	32.0	36.1	47.2	33.2	46.8	5.5	100	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	72.2	61.0	69.5	60.1	66.6	58.7	62.2	69.5	60.9	67.6	59.7	65.6	8.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	74.8	64.2	57.3	70.3	63.2	56.4	68.0	62.6	27.7	69.5	26.9	67.6	26.1	65.8	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.5	26.3	22.8	DB	18.8	79.5	4.5	3.4	15.5	81.9	12.9	83.9	10.4	85.8	7.2	88.3	(4)
(5)			WB	17.1	65.1	4.5	1.4	13.8	66.2	11.2	67.0	8.7	67.8	5.4	68.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	42.2	42.3	45.6	48.9	52.9	56.8	59.4	59.8	56.8	51.0	45.7	41.4		
	DBStd	8.02	6.20	5.24	4.63	3.75	3.59	3.22	3.32	3.21	3.78	4.56	6.10	6.01		
	HDD50	1154	247	218	149	61	13	0	0	0	1	42	152	271		
	HDD65	5385	707	635	600	482	375	248	177	166	247	435	580	733		
	CDD50	1255	5	3	14	30	103	203	292	303	205	72	22	3		
	CDD65	10	0	0	0	0	0	1	4	4	1	0	0	0		
	CDH74	37	0	0	1	0	2	4	17	10	3	0	0	0		
	CDH80	2	0	0	0	0	1	0	1	0	0	0	0	0		
(14) Wind	WSAvg	8.4	9.9	9.2	10.2	9.0	7.5	7.2	6.6	5.9	5.7	8.1	10.5	10.7		
(15) Precipitation	PrecAvg	19.5	2.4	1.6	1.6	1.5	1.3	1.1	0.7	0.9	1.2	1.7	2.6	2.6		
	PrecMax	30.6	6.1	3.3	3.6	2.6	2.7	2.7	2.1	2.1	4.2	5.8	8.9	6.5		
	PrecMin	13.6	0.4	0.6	0.3	0.4	0.4	0.0	0.0	0.0	0.0	0.3	0.7	0.4		
	PrecStd	3.8	1.3	0.6	0.7	0.6	0.6	0.7	0.6	0.7	1.0	1.1	1.6	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	57.4	62.9	64.1	67.5	73.2	76.0	75.2	72.1	66.3	63.2	56.9		
		MCWB	51.4	50.1	52.3	54.0	57.5	61.0	62.9	61.8	60.5	57.2	55.4	51.0		
	2%	DB	54.3	54.2	57.3	60.4	63.7	68.5	72.0	71.7	68.3	63.1	58.2	53.3		
		MCWB	49.9	48.5	50.1	51.7	55.9	59.1	61.2	61.2	59.2	56.2	53.6	49.3		
	5%	DB	52.3	52.0	54.8	57.4	61.5	64.8	68.6	68.6	65.9	60.5	55.7	51.3		
		MCWB	48.1	47.4	48.8	50.4	54.7	57.3	60.0	60.0	58.3	54.7	51.5	47.2		
	10%	DB	50.3	49.8	52.8	55.3	59.4	62.9	65.8	66.1	63.7	58.0	53.9	49.4		
		MCWB	46.5	45.9	47.7	49.5	53.5	56.2	58.6	59.0	57.3	53.2	50.1	45.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	52.3	54.1	55.4	59.4	62.4	64.2	63.6	62.7	60.1	57.0	52.7		
		MCDB	55.8	55.4	59.4	61.7	64.2	70.0	73.8	70.7	68.3	64.3	61.2	55.4		
	2%	WB	51.0	50.0	51.4	53.3	57.1	60.0	62.1	62.0	60.5	57.4	54.4	49.9		
		MCDB	53.2	52.9	55.3	58.1	62.5	66.9	70.3	69.0	65.9	61.1	57.5	52.5		
	5%	WB	48.8	47.9	49.9	51.6	55.4	58.2	60.4	60.7	59.3	55.4	52.6	47.9		
		MCDB	51.4	50.9	54.0	56.1	60.5	64.0	67.4	67.2	64.3	59.1	55.5	50.6		
	10%	WB	46.9	46.3	48.2	50.4	54.1	56.8	59.1	59.6	58.0	54.0	50.4	46.2		
		MCDB	49.4	48.8	52.1	54.8	58.6	62.0	65.2	65.4	62.9	57.2	53.3	48.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	11.7	12.3	12.4	12.0	12.2	12.9	13.3	13.6	12.3	10.6	9.3		
		MCDBR	10.5	13.3	14.7	15.7	14.9	16.1	17.4	16.9	16.2	14.0	10.5	9.1		
		MCWBR	7.5	9.1	9.3	9.4	9.2	9.1	9.1	8.9	9.2	8.9	7.3	7.0		
	5% WB	MCDBR	9.2	10.8	13.0	14.5	14.0	15.0	16.3	15.4	14.1	11.5	9.3	8.0		
		MCWBR	7.3	8.2	9.0	9.3	9.2	9.4	9.1	8.6	8.6	8.3	7.3	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.314	0.327	0.349	0.372	0.355	0.346	0.351	0.338	0.328	0.320	0.313		
	taud		2.536	2.560	2.540	2.469	2.417	2.486	2.539	2.527	2.558	2.576	2.537	2.502		
	Ebn at Noon		242	272	285	287	282	288	289	283	277	262	238	227		
	Edh at Noon		19	23	27	32	35	33	31	30	27	22	19	18		
(44) All-Sky Solar Radiation	RadAvg		297	577	898	1354	1730	1891	2070	1749	1249	689	355	239		
	RadStd		40	77	105	126	136	180	146	114	104	75	53	32		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+4
Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page